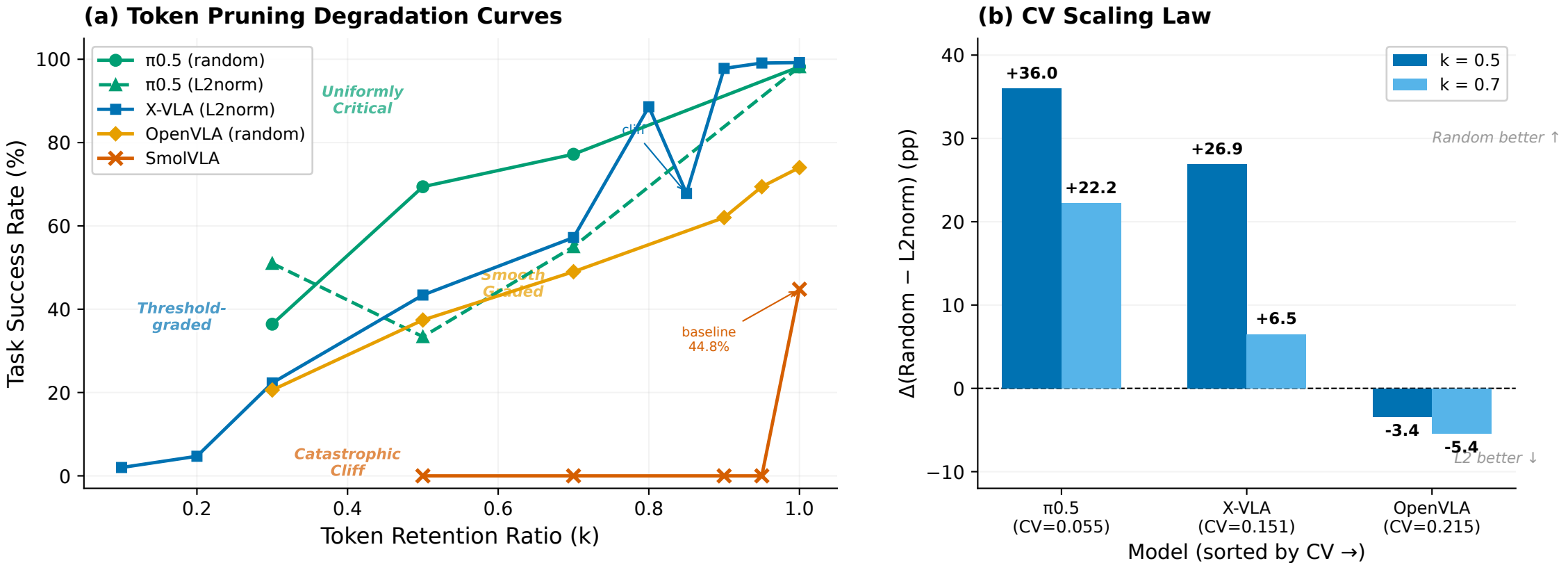
(a) Token Pruning Degradation Curves
π0.5 (random)
π0.5 (L2norm)
X-VLA (L2norm)
OpenVLA (random)
SmolVLA
Uniformly Critical
cliff
Threshold-graded
Smooth Graded
baseline 44.8%
Catastrophic Cliff
Task Success Rate (%)
Token Retention Ratio (k)
(b) CV Scaling Law
k = 0.5
k = 0.7
+36.0
+22.2
+26.9
+6.5
-3.4
-5.4
Random better ↑
L2 better ↓
Δ(Random − L2norm) (pp)
π0.5 (CV=0.055)
X-VLA (CV=0.151)
OpenVLA (CV=0.215)
Model (sorted by CV →)